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Being Responsibly Digital: A Care Ethics Lens for Digital Responsibility

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1 Introduction

Digital technologies permeate every aspect of our personal and professional lives. Artificial intelligence (AI), for instance, is integrated into decision-making systems across sectors, from healthcare to manufacturing; data-intensive infrastructures sustain functions from transportation to education; digital platforms increasingly mediate public discourse, intimacy, and labour. Such a profound digitalisation of the very fabric of social, personal and professional worlds has brought to the fore the impetus for responsibility in research and practice. Responsibility, here, as a term, can be interpreted and applied in various ways. Within the context of research and innovation, for instance, responsibility entails anticipating, reflecting, and responding to societal needs while adopting an inclusive practice (Stilgoe et al., 2013). Through such an approach, it is argued that the outputs and outcomes of research and innovation are more likely to be both needed and desirable (Stahl, 2013). Digital responsibility, thus, may similarly encompass such a focus on pursuing positive and desirable outcomes, while considering the accountability for doing so and the commitment to pursue this consistently (Recker et al., 2025).

Despite the proliferation of guidelines, frameworks (e.g., UKRI, 2023; Wilford et al., 2016) and governmental initiatives (e.g., DESIT, 2024), however, within policy and industry, digitalisation is often viewed exclusively through a positive lens. This is to an extent understandable, particularly considering that digitalisation is often underpinned by needs and desires that centre around efficiencies, market competition and productivity gains (Zamani & Vannini, 2025). Such a perspective, however, can be problematic because technological optimism tends to downplay potential negative consequences, which for many may be subtle but are significant, especially for those most underserved (e.g., Rousaki et al., 2024; Zamani & Rousaki, 2024). Digital technologies can generate and amplify relational inequalities (Zheng & Walsham, 2021), as data flows and practices create dependencies and power asymmetries (Angelopoulos et al., 2021; Heeks, 2022). Algorithmic systems, in particular, shape and influence opportunities and freedoms, while further disempowering vulnerable individuals and communities that cannot be easily contested (Kayas et al., 2019). Despite the attention on digital responsibility, however, harms persist, vulnerabilities multiply and the divide between aspirations for responsible digital technologies and the lived experiences of digitalisation outcomes continues to grow.

In this editorial, we argue that digital responsibility needs to be viewed through a care ethic lens. While existing approaches often frame responsibility as a series of obligations, such as mitigating risks, being inclusive and reflective of one's practices and of the research and innovation process, ethics of care emphasises relationality, interdependence, contextual judgment, and attentiveness to vulnerability (Geiger et al., 2024). Approaching digital responsibility through care ethics foregrounds duties of responsiveness, maintenance, repair, and long-term engagement (Tronto, 1998). By rejecting the premise that people encounter digital technologies as autonomous agents who freely choose if and how to engage with them, care ethics sheds light instead into the relationships between, for instance, users and designers, and platforms and their communities. By drawing attention to these relational dynamics, digital technologies are considered as systems of power that expose individuals to surveillance, discrimination and exclusion; for instance when people rely on them for accessing health and social care (Rousaki et al., 2024), drawing attention

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to those who bear the burdens of digitalisation, and those whose voices are often excluded from technological futures.

In what follows, we develop the argument for why care ethics is indispensable for contemporary debates about digital responsibility. We begin by examining the shifting landscape of digital responsibility and the limitations of dominant frameworks. We then articulate the conceptual foundations of care ethics and show why it is particularly well suited to address the ethical challenges of accelerated digitalisation. Finally, we introduce the contributions of the Special Issue 'Being Responsibly Digital' and discuss broader implications for research, policy, and practice. Together, these interventions highlight why caring, rather than being considered as a peripheral concept, should be instead central to how we imagine and enact responsibility in our digital lives and workflows.

2 Digital Responsibility within an Ever-changing Landscape

2.1 Prevailing Models of (Digital) Responsibility

The concept of responsibility has evolved significantly over the past decade. Early discussions primarily emerged from fields such as computer science (e.g., Beusmans & Wieckert, 1989), corporate governance (e.g., Lantos, 2002) and innovation (e.g., Kiran, 2012). Within these early conversations, responsibility has been typically framed as compliance, or adhering to ethical guidelines and procedures, primarily with a focus on mitigating identifiable harms. Over the years, and as the discourse matured, conversations on responsible innovation and research, often inspired by Science and Technology Studies (STS) research, introduced more expanded and in-depth approaches to responsibility, bringing into the fore anticipatory thinking, commitment to stakeholder engagement and the principles of value-sensitive design into technology development and organisational processes (Stilgoe et al., 2013). As Stahl (2012, p. 208) explains, these conversations, implicitly or explicitly, encourage us to think about and engage in "technology foresight with an awareness of the necessity to engage in broader social discussions", which further suggests that responsibility requires being both willing and able 'to be responsible'.

When operationalising the above in practical terms, research suggests that several dimensions need to be considered, namely: diversity and inclusion; anticipation; reflexivity and deliberation; responsiveness and adaptative change; openness and transparency; and sustainability (Kwee et al., 2021). These dimensions have also been acknowledged as crucial components for digital responsibility in particular (e.g., Trier et al. (2023) propose very similar principles such as participation, accountability, privacy, sustainability etc.). These dimensions are critical components for the design and development of any research project, technological product or information system. We further contend that within the context of accelerated digitalisation, they become even more significant in terms of how people, be it as end-users, technologists, researchers, managers or decision-makers, think about the digital world, and engage with it.

2.2 On the Limitations of Traditional Approaches to Digital Responsibility

It is at this juncture that we observe certain shortcomings precisely in the way these dimensions (or principles) become operationalised within larger frameworks. For example, rights-based approaches that focus on protections against possible harms, especially due to digital technologies, become significantly weaker within the context of our contemporary political-economic condition where digital platforms and infrastructures concentrate ownership, control, and rents in the hands of a few powerful actors (Saura García, 2024). Similarly, the adoption of risk-based approaches, and which rely on calculability and predictability, become increasingly complex and difficult, and often disregard structural injustices and human rights (Zamani & Rousaki, 2026). Principle-based approaches, while they support judgment and deliberation (Porter et al., 2024), because they emphasise universal norms, they risk being too abstract to be meaningful in ever diverse contexts and across borders (Munn, 2023), and may clash with other approaches (e.g., rules-based) (Sama & Shoaf, 2005).

Finally, positioning these within a framework of digital responsibility cannot sufficiently address the implications and potential harms of continuous and accelerated digitalisation, whereby sociotechnical concerns, ethical and regulatory frameworks become entangled with digital technologies at multiple levels, in unexpected ways and where considering and mitigating against possible unintended consequences may not be enough (Recker et al., 2025). Here, we offer some examples. As data flows increase through AI-enabled systems and platforms, new forms of governance emerge and make responsibility harder to locate due to the ever-increasing complexity of systems (Slota et al., 2023). The speed and scale of digitalisation also produce new asymmetries between those who design technologies and those affected by them through direct or indirect use, and even non-use; between platforms that benefit from data and individuals whose lives become datafied (Tironi & Valderrama, 2022). The effects of digital technologies may unfold over time in predictable and less predictable ways, with harms being direct and immediate or indirect and present themselves much later (Saghafi & Angelopoulos, 2025).

These arguments suggest that digital responsibility cannot be solely anticipatory, principles- or rules-based. Rather, it requires continuous attention to the emergent and lived effects of digital infrastructure, precisely because the lived reality of continuous digitalisation resists such simplification: digital technologies are entangled with social norms, political structures, environmental infrastructures, and everyday practices. They operate through algorithms and datasets as well as through formal and informal decision-making, organisational cultures, historical inequities and structural injustices, power dynamics and market logics. Traditional approaches to digital responsibility frequently fail to engage with these relational, situated, and systemic aspects as they tend to presume that ethical dilemmas can be resolved, for instance, via deliberation, transparency or the application of commonly agreed principles and rules. As De Vaujany et al. (2025) argue, digital responsibility needs to be consider as an ongoing, relational practice that takes place over time rather than as a static set of principles.

Care ethics offers such an approach and further enriches it by acknowledging the dependencies created by digital technologies, the great range and diversity of what vulnerability might mean, and the need for a sustained engagement throughout, whereby maintenance, repair and accountability are valued as much as innovation, efficiencies and speed. In addition, and as we will show later, care ethics offers the required conceptual tools for analysing how commitments are enacted, distributed, and sustained within complex social and institutional arrangements. These two taken together make care ethics particularly well suited for engaging with the challenges of highly digitalised, and often automated environments, where responsibility is often diffused and deferred, and where the impacts of accelerated digitalisation are uneven and extend across time and sectors.

In what follows, therefore, we turn our attention to care ethics that provides an indispensable tool for grounding digital responsibility in relationality, interdependence, and sustained attentiveness to vulnerability (Tronto, 1998, 2010).

3 Care Ethics as a Conceptual Lens for Digital Responsibility

3.1 Theoretical Foundations of Care Ethics

Care ethics is an ethical model that finds its origins in moral development (Gilligan, 1993) and care-focused feminism (Noddings, 2013). It is a distinctive ethical framework that challenges dominant, principle-based and individualistic accounts of moral responsibility by putting forward the notion of relationality, interdependence and situated judgement. In other words, care ethics understands responsibility as emerging through the confluence of social, material and institutional relations (Xu & Smyth, 2023), and concrete relationships of dependency, power, and vulnerability (Sevenhuijsen, 2003). The main concern is not whether principles are universally and consistently respected, but rather whether and how our everyday practices can produce and sustain the conditions that can allow people and their communities to flourish.

Attentiveness to others' needs is at the core of care ethics. It is when needs and vulnerabilities remain invisible or become normalised, that we observe moral failure, exclusions and harms (Heath et al., 2025). Here, being attentive is not a passive state; it requires contextual sensitivity and openness to other's experiences and perspectives that may challenge one's priorities and assumptions, because it is through this that we can recognise a need and respond to it. Yet, Tronto (1998) argues that being aware and paying attention to the need for caring (*caring about*) is not sufficient; rather, it is required that somebody assumes the responsibility for caring (*caring for*), where concerns need to translate into commitment for taking action. This should not be confused with obligation, and the difference between responsibility and obligation is central to care ethics. Whereas obligation reflects duties that are assigned or prescribed through generalisable rules and universal principles that are context-agnostic, responsibility is relational, context-specific and sensitive to dependency and inequalities (Tronto, 2005). In other words, responsibility must be explicitly assumed rather than tacitly displaced or deferred.

Beyond the assumption of responsibility, care ethics emphasises the importance of competent action, whereby caregiving requires both "perform[ing] the necessary caring tasks [and i]t involves knowledge about how to care" (Tronto, 1998, p. 17), and thus requires capacity and competence, resources and supportive structures. This draws attention to the too often invisible labour involved in maintaining and repairing our world through responding to others' needs (Fisher & Tronto, 2003), where care takes place over time and requires our ongoing engagement rather than one-off interventions. This is crucial because, this ethical framework goes beyond identifying, committing to and explicitly addressing others' needs, but it further necessitates a bidirectional relationship and closing the loop through feedback on and evaluation of the care received. It is this stage that allows us to improve and adapt our caregiving when it falls short, to avoid paternalism and to meaningfully recognise one's agency and their lived experience (Sevenhuijsen, 2000).

3.2 What an Ethic of Care Contributes to the Digital Responsibility Discourse?

Traditional digital responsibility frameworks and approaches have been instrumental in extending conceptualisations of responsibility beyond just compliance and risk management. Yet, they are still largely oriented, as critics argue, toward governance and institutional processes, often lacking a focus on the political and social relations and structures that influence empowerment and emancipation (Penttilä, 2024) within digital spaces, and even risking responsibility practices becoming "empty signifiers" that perpetuate inequalities (Felt et al., 2023, p. 18). Here, approaching digital responsibility through a lens of care ethics allows the dimensions commonly associated with responsibility (i.e., inclusion, anticipation, reflexivity etc.) to be reinterpreted and applied as relational and ongoing commitments rather than as purely procedural requirements. This is because care ethics embeds these dimensions within an ontology of interdependence, vulnerability and responsibility over time. In what follows, we unpack our argument, tracing the phases of caring and mapping them against the traditional digital responsibility dimensions.

First, '*caring about*' emerges through a moral quality of attentiveness. It suggests that stakeholder engagement is not simply an exercise of representation and balance, but rather a requirement to think whose vulnerabilities, experiences and needs might become invisible in or harmed by digital systems. In this sense, digital exclusion is not a matter of, for instance, a skills deficit, but a matter of failure to care about underserved individuals and communities in the first place (Rousaki et al., 2024; Zamani & Rousaki, 2024; Zamani & Vannini, 2025). Therefore, '*caring about*' goes beyond the procedural inclusion of traditional approaches to digital responsibility by asking who defines relevance, whose harms count as harms, and which forms of exclusion remain normalised within digital infrastructures. More importantly, care ethics reframes anticipation of harms and unintended consequences: rather than thinking digital responsibility in terms of foresight and risk assessment, anticipation shifts from a predictive exercise to a sustained ethical orientation toward those who may be disproportionately affected by digital transformation.

Second, '*caring for*' addresses a key shortcoming of traditional approaches because it entails that besides identifying risks and ethical concerns, it necessitates assuming the responsibility of addressing them. In other words, reflexivity and deliberation are not the final objectives but instead they become mechanisms for taking up or allocating responsibility and considering what are the required conditions and resources. We also posit that '*caring for*' also deepens our engagement with the requirements for openness and transparency, typically present in the digital responsibility discourse (e.g., Recker et al., 2025). This is because from a care ethics perspective, '*caring for*' frames transparency not as a matter of information disclosure, but a relational requirement that enables responsibility to be assumed and contested, which is crucial for enabling those impacted to appreciate who is responsible and challenge them.

Third, *caregiving* itself aligns with being responsive and reflects the need for adaptive change. *Caregiving* suggests that responsiveness goes beyond (institutional or individual) capacity to adjust processes and outputs, because it explicitly requires that actors need to act competently in response to real-world harms and impacts, including to unintended consequences. This transforms adaptive change into a core responsibility grounded in the lived experiences of those affected by digital systems. We observe that caregiving can also extend our understanding of what sustainability means. As Brown (2025, p. iii) argues, within the Information Systems discipline and the digital responsibility discourse, "sustainability is not confined to organizational "green" strategies or corporate carbon targets alone. It extends to how we design, use, govern, and theorize about digital technologies, such that the people and organizations can reap the benefits and not be harmed in the process". Caregiving contributes toward this by foregrounding the labour of maintenance, repair and social relations (Alacovska & Bissonnette, 2021), and thus highlights valuing and resourcing responsibility work rather than prioritising continuous innovation and growth alone.

Fourth, *care-receiving* introduces a dimension largely absent from existing digital responsibility frameworks by focusing precisely on the lived experience of those affected by digital systems and digitalisation as a whole. Penttilä (2024, p. 5) highlights that traditional responsibility frameworks are often unable "to go beyond an (expert) multidisciplinary approach and to incorporate a broader political dimension" and that this results in essentially legitimising "the exclusion of the public in deliberative and participatory practices". Here, care-receiving operationalises responsiveness and adaptive change through feedback, contestation, and revision grounded in the lived experiences of those directly or indirectly impacted. In doing so, it resists paternalistic assumptions that responsible action can be fully specified in advance and insists that responsibility needs to remain open to scrutiny and contestation by those with lived experience rather than purely by experts, and that their contributions is equally valued.

Table 1 summarises the above by mapping care ethics against the digital responsibility dimensions.

4 Being Responsibly Digital

Looking at digital responsibility from a care ethics lens suggests that it requires more than compliance with principles

Table 1 Mapping of Care Ethics against Digital Responsibility Dimensions

Care Ethic Phases	Digital Responsibility Dimensions
Caring about: being attentive	Attentiveness to diverse needs, vulnerabilities, and emerging social impacts; anticipation and awareness of potential impacts and harms
Caring for: assuming responsibility	Explicit assumption and allocation of responsibility through reflexive deliberation, accountability, and relational transparency
Caregiving: acting competently	Competent and responsive action, adaptive change grounded in long-term sustainability, maintenance, and stewardship
Care-receiving: responding to care	Ongoing evaluation through feedback, learning, and adaptation based on lived experience of those affected

and regulations, and foresight for mitigating possible risks. Rather, being responsibly digital calls for an orientation that formally recognises that digital systems as relational infrastructures that impose and organise dependencies, (re)distribute power and influence and shape the conditions within which individuals and communities live and act. Implicitly, this requires that digital responsibility does not begin at the point of design choices and does not conclude with being accountable alone either. Care ethics requires, instead, that digital responsibility needs to take effect much earlier in the process by being explicitly attentive to whose needs our digital systems aim to respond to or potentially ignore and who becomes visible or invisible through digitalisation. Importantly, this goes beyond recognising that digital technologies can have both positive and negative outcomes (e.g., Recker et al., 2025), because care ethics explicitly goes beyond impacts to consider needs and vulnerabilities (Bendl et al., 2019; Fotaki et al., 2019). In other words, care ethics draws attention to power structures and politicises attentiveness by asking whose experiences are counted as relevant, and whose harms may become normalised.

To be responsibly digital is, therefore, to resist the assumption that people engage with technology as isolated, autonomous agents and instead to acknowledge that digital technologies are embedded in social, institutional, and political relations that produce uneven impacts over time. The care ethics lens enhances this understanding by helping us frame responsibility as an ongoing practice rather than a static exercise. Drawing on Tronto, being responsibly digital entails promoting attentiveness to emerging dependencies and cumulative harms, an explicit assumption and fair distribution of responsibility for addressing those needs, and it then demands that competent, responsive action is taken up and sustained as digital systems evolve to maintain and repair our digital world. Crucially though, responsibility cannot be claimed without opening up to the feedback from those affected and a willingness to adapt digital systems when these fail them or cause harms. In this sense, responsible digitalisation is not exhausted by anticipation or governance mechanisms; it is realised through responsiveness, learning, and sustained ethical engagement. The above are illustrated in Fig. 1.

5 Contributions of this Special Issue

In this special issue, we were interested in studies of digital responsibility within a broad spectrum of technologies, emerging, disruptive and more conventional ones. We were also interested in studies that can help us understand better what digital responsibility might mean within diverse contexts. Eventually, and following the review process, nine articles were selected to be included in the special issue. Each of these articles focuses on different domains and adopts diverse methodological traditions. Collectively, however, they help us advance our understanding on the implications, impacts and outcomes of digital technologies and systems, and further theorise different aspects of what being responsibly digital means.

As advanced technologies become increasingly autonomous and capable of self-learning, the nature of human-AI interaction is also changing, bringing new ethical challenges to the forefront. Passlack et al. (2025) argue in their article for an expanded understanding of AI literacy, that goes beyond technical proficiency to further include the skills required for responsible engagement with AI systems. Rather than focusing solely on efficiency and performance, they adopt a digital responsibility perspective that addresses potential risks and unintended consequences and identify key factors that shape and enable responsible interaction with AI technologies. Their findings further reveal the underlying tensions in how AI literacy is currently understood, particularly around skill development and the attribution of responsibility. These tensions highlight the importance of shared responsibility and of aligning individuals' capabilities with their professional roles.

Still within the realm of AI, Akbarighatar et al. (2025) discuss the development, deployment, and governance of responsible AI (RAI) through the perspectives of experts. The authors note that RAI refers to the development, deployment, and governance of AI systems which are human-centred, trustworthy, and aligned with societal values. The study focuses on the guiding principles that organisations apply with the intention to reduce risks such as bias and privacy infringements while promoting transparency, fairness, and other desirable outcomes. Using a sequential mixed-methods design, the authors uncover five distinct combinations of RAI principles that are associated with high levels of overall RAI enactment, further highlighting the significance of accountability and safety–reliability, and drawing

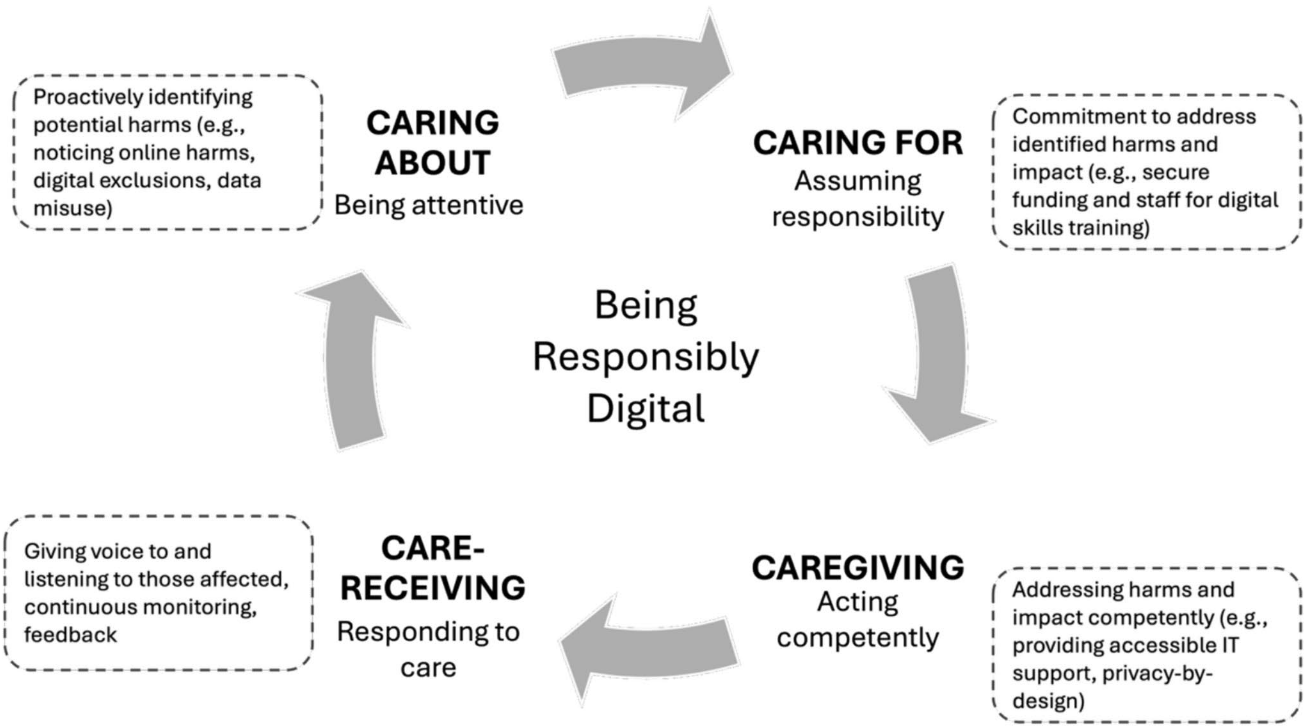

Fig. 1 Being Responsibly Digital Framework

attention to the complex interdependencies and contextual conditions that shape RAI practices.

Another AI-focused study is that by Olan et al. (2025). The authors focus on explainable AI (XAI) and draw attention to how it is currently reshaping social entrepreneurship, marketing, and service activities. They are primarily interested in how and under what conditions B2B marketing activities support social entrepreneurship, and they thus examine how XAI affects B2B service organisations. Drawing on theories of social entrepreneurship, B2B marketing, and emerging technologies, the study identifies and conceptualizes key factors and their interrelationships, and leverages survey data to demonstrate that XAI supports the development of sustainable resilience in B2B marketing activities and contributes to the formulation of social entrepreneurial strategies that drive B2B marketing innovation.

Besides AI-focused studies, however, the special issue contains articles that move away from this technology. In their paper, Currie and Seddon (2025) focus on the Bitcoin and examine elements and matters pertaining to adoption, trust, and regulation, while sharply focused on the dialectical debates and competing narratives currently forming the discourse at the intersection of Bitcoin as a digital asset and digital responsibility. As they show through their findings, tracing digital responsibility spans the move from transformation to adoption.

With sustainability being today an increasingly urgent global priority, organizations are turning greater attention to the role of IS in supporting environmental goals. Nieskens and Saghafi (2025) focus on this domain, and explore how organisational and societal influences shape IT professionals' beliefs about environmental sustainability and how these beliefs, in turn, affect their intentions to engage in pro-environmental behaviours. Specifically, they examine the above dynamics at the individual level, where empirical evidence is still limited. Their work uncovers key drivers of Green IT adoption intentions. The findings shed light on the antecedents of sustainability-related beliefs and pro-environmental attitudes among IT professionals, offering meaningful insights for both scholars and practitioners.

Kumar and Shirish (2025) focus on employee advocacy programs and digital workplace platforms aimed at mitigating negative well-being outcomes such as isolation and disengagement, and at encouraging also proactive organizational citizenship behaviours (e.g., digital employee advocacy). Their work evaluates the well-being and social implications embedded in the design intentions of employee advocacy platforms through a bricolage approach of user reviews of these platforms. They further identify crucial and relevant dimensions and develop a well-being–driven grounded model that connects platform-enabled social and psychological well-being to digital employee advocacy behaviours through mechanisms of social and emotional attachment. The authors further contribute to the digital responsibility agenda and this special issue by advancing our understanding of digital responsibility enactment and digital employee advocacy as a key outcome.

Within the healthcare sector and the context of its accelerated digital transformation, Urban and Plattfaut (2025) focus on the unintended and potentially harmful consequences of widespread digital technology use. In this context, the notion of acting responsibly in digital environments has gained increasing importance and has drawn our attention to the need for deeper conceptual clarity around digital responsibility and a better understanding of its role within digital transformation initiatives. The authors examine the relationship between digital responsibility and digital transformation in the setting of large-scale transformation efforts and make a significant contribute to the digital responsibility agenda by identifying the dynamic interplays between digital responsibility and digital transformation.

Lorini et al. (2025) contextualise their study around vulnerable migrant populations to approach digital responsibility as the ethical and accountable design and use of digital technologies. Developing the knowledge and skills needed to engage with the digital world safely, effectively, and securely, the authors note, is particularly crucial in contexts where digital literacy and access are limited and livelihoods are fragile. Using the AREA Plus framework as their analytical lens, they present the findings of two projects based on co-creation where digital interventions aimed at improving their living conditions. In doing so, they enrich the AREA Plus by introducing sustainability as an additional dimension. Among the study's contribution, we highlight the detailed procedures for conducting research with vulnerable populations leveraging co-creation as core part of the process.

Finally, Panteli et al. (2025) argue for the growing need to examine the scale and scope of responsible cybersecurity, which is of paramount importance due to our increasingly connected digital world. In doing so, they conducted interviews with senior cybersecurity professionals to explore what responsible cybersecurity might mean and what forms it might take. The findings of their work showcase that responsible cybersecurity spans techno-centric, human-centric, organisational and societal centric perspectives. In addition, the authors highlight that these different perspectives are linked up to the responsibilities of various stakeholders and further highlight the crucial role of senior leadership in this context.

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Data Availability We do not analyse or generate any datasets, because our work proceeds with a theoretical approach.

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